

FELIPE DIOGO

Braga – Portugal

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🌐 0xfelipegd.github.io

Summary

Automation and Control Engineer with 6+ years of hands-on experience who has delivered a 1,200 kg UGV and a production-grade IIoT monitoring platform from scratch — while also wiring industrial panels, programming PLCs, and leading engineering teams. Deep expertise across the full Industry 4.0 stack: field-level devices, industrial protocols, IT/OT convergence, edge computing, and cloud data pipelines. Equally comfortable in a control cabinet or a code editor. Founder of Movewer Technologies; open to senior automation, robotics, or embedded-systems engineering roles across Europe.

Education

Federal Institute of São Paulo

Bachelor of Control and Automation Engineering

Jan. 2019 – Dec. 2023

São José dos Campos, São Paulo

Relevant Coursework

Hands-on, project-based coursework — built and wired real systems in the lab, not just theoretical study.

- PLC and SCADA Programming
- Industrial Networks & Fieldbuses
- 3D Modeling and CAD Software
- Web Development
- IT/OT Convergence
- Product Development
- Python and Node.js Development
- Electrical Panel Design & Wiring
- IoT, MQTT and Industry 4.0

Technical Skills

Automation & Industry 4.0: PLC programming (TIA Portal, CODESYS, GX Works), SCADA/HMI (Indusoft, WinCC), industrial protocols (Modbus RTU/TCP, Profinet, EtherCAT, OPC-UA, LoRa), electrical panel design & wiring, IT/OT convergence, IIoT architecture, edge computing, control systems, power electronics, embedded systems

Software & Data: Python, JavaScript/TypeScript, Node.js, Node-RED, MQTT (Mosquitto), InfluxDB, Grafana, N8N, PowerBI, Git/GitHub, Linux, web development, AI integrations, cloud deployment

Engineering Tools: SolidWorks, EPLAN, AutoCAD Electrical, MATLAB, Eagle, Raspberry Pi, NVIDIA Jetson Nano/Orin

Management: Scrum, Kanban, Jira, version control, cross-functional team leadership, budget and schedule management

Experience

Movewer Technologies

Founder & Automation Engineer

Jun. 2025 – Present

Portugal

- Architected end-to-end Industry 4.0 solutions for UGV (Unmanned Ground Vehicle) platforms, covering control systems, power electronics, wireless communication, remote-control stations, and multi-sensor integration.
- Built a full-stack Industrial IoT Monitoring Platform using MQTT, Python, Node-RED, InfluxDB, and Grafana; integrated AI-driven analytics to surface actionable insights beyond raw sensor data.
- Defined IT/OT convergence architecture, bridging field-level devices to cloud dashboards with real-time data pipelines and automated alerting.
- Collaborated with a mechanical specialist to co-design modular UGV systems aligned with client requirements and safety standards.

Psyche AeroSpace

Automation Executive Manager

Jun. 2024 – Jun. 2025

São Paulo, Brazil

- Built the Automation and UGV divisions from zero; systems delivered contributed directly to a new fundraising round, with technical demonstrations cited as key proof points by company leadership.

- Led a cross-functional team of 5 engineers across two product lines (agricultural UGVs and drone-support infrastructure), reducing time-to-prototype by ~30% through structured Scrum sprints.
- Owned the full development lifecycle of UGV systems—control, power, and communication—coordinating with mechanical and electronics teams across tight delivery timelines.
- Delivered a 10 kg-payload agricultural quadcopter from concept to flight-ready prototype in under 6 months, including electrical project, mechanical integration, and flight-control and telemetry systems.

Controvale

Jun. 2018 – Jun. 2024

Automation Engineer

São Paulo, Brazil

- Engineered 20+ special industrial machines integrating mechatronic systems, IoT connectivity, SCADA supervision, and industrial communication protocols (Modbus, Profinet, EtherCAT); average project delivered on time and within a 5% budget variance.
- Designed and executed full industrial electrical panel projects end-to-end — load calculations, schematics (EPLAN/AutoCAD), component specification, physical assembly, commissioning, and field testing; reduced rework rate by ~25% through standardized documentation practices.
- Programmed PLCs and configured HMI/SCADA systems (TIA Portal, CODESYS, Indusoft) for manufacturing and process automation projects, improving operator visibility and reducing unplanned downtime across client facilities.
- Served as regional distributor for Siemens, Rockwell, Bosch, Wecon, and Kinco; built and maintained a portfolio of 35+ industrial client partnerships over 6 years.

Projects

MOV PLATFORM | *Industry 4.0, MQTT, IoT, InfluxDB, Node-RED, Grafana, AI Analytics, VPS*

Jan. 2026

- Production-grade IIoT Monitoring Platform with AI-powered insights deployed on a VPS. Edge layer: Node-RED on Raspberry Pi → MQTT broker → InfluxDB time-series storage → Grafana dashboards with anomaly detection and trend forecasting. [0xfelipegd.github.io](https://github.com/0xfelipegd)

HIGH-PAYLOAD UGV – ARMAX | *Autonomous Ground Vehicle, Power & Control Systems, Modular Design*

Aug. 2025

- Full development of a 1,200 kg-payload UGV. Build-ready deliverable includes system architecture, power-electronics design, wireless control stack, performance benchmarks, and commercial feasibility analysis. movewertechnologies.com

AGRO DRONE | *Quadcopter, Medium-Payload UAV, Flight Control & Telemetry*

Apr. 2025

- Medium-payload agricultural quadcopter from requirements to flight-ready prototype: frame selection, ESC/motor sizing, flight-controller tuning, and telemetry integration—on schedule and within budget. [0xfelipegd.github.io](https://github.com/0xfelipegd)

DRONE ASSISTANCE BASE | *Automated Process Plant, Energy Management, Intelligent Control Hub*

Dec. 2024

- Automated ground-support complex for high-payload agricultural drones: autonomous refueling, energy management, and a centralized intelligent control hub with fully automated field-intelligence-driven workflows. [0xfelipegd.github.io](https://github.com/0xfelipegd)

Languages

Portuguese — Native **English** — Advanced (B2/C1)

Extracurricular

Dual-Track Professional

Automation Engineer & Software Developer

- Pursuing parallel careers in industrial automation/robotics and software development, applying both disciplines to build integrated Industry 4.0 products.
- Independently developing web applications and automation tools, with a focus on IIoT dashboards and data-driven manufacturing solutions.

Startup Founder

Movewer Technologies

- Founded a technology startup focused on Industry 4.0 solutions and unmanned ground vehicles (UGVs), delivering hardware and software products to industrial and agricultural clients.